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Mathematical Scale Restoration: Parameter-Free Retrieval Augmentation for Time Series Foundation Models

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Abstract Time Series Foundation Models (TSFMs) promise zero-shot forecasting, yet recent retrieval-augmented generation (RAG) approaches rely on expensive, dataset-specific neural adapters to fuse historical context. While neural mixers natively handle scale variations in dense, continuous data, their computational overhead breaks the zero-shot paradigm. Conversely, parameter-free statistical retrieval (k-NN) suffers from catastrophic scale mismatch when historical continuations differ in magnitude from the query sequence. In this paper, we propose *ScaleRAG*, a universal mathematical framework whose retrieval and scale-restoration stages are entirely parameter-free: explicit scale normalization prior to indexing and exact statistical scale restoration post-retrieval, with no learned projector and no neural encoder. Through rigorous ablation across both sparse retail (M5) and dense sensor (ETTm2) regimes, we show that explicit restoration prevents a $3.8\times$ degradation in RMSSE on sparse data and recovers retrieval quality by +85.4% across modalities. On the dense ETTm2 benchmark the restored candidate is fused with the frozen backbone through a fixed, fully parameter-free weight; on the sparse M5 panel we add a single lightweight learned component—a LightGBM gate over six diagnostic features, fitted on training origins only—which is the only trainable part of the pipeline. Furthermore, we define a regime taxonomy: neural RAG dominates continuous datasets, while explicitly restored statistical retrieval yields its largest gains in intermittent, scale-heterogeneous regimes. We report this boundary honestly—on the held-out M5 test panel ScaleRAG improves squared-error accuracy over its frozen backbone but does not meet our pre-registered improvement threshold over strong classical baselines and does not win the official WRMSSE—and argue that the contribution is a validated mechanism and regime taxonomy rather than a state-of-the-art accuracy claim.

Keywords Time Series Forecasting · Retrieval-Augmented Generation · Foundation Models · Zero-Shot Learning · Scale Invariance

1 Introduction

The emergence of Time Series Foundation Models (TSFMs) has shifted forecasting paradigms from dataset-specific training toward zero-shot generalization. Models such as Chronos [1], Moirai [20], and Lag-Llama [15] have demonstrated remarkable ability to infer temporal patterns across diverse domains. However, standard zero-shot TSFMs struggle to synthesize information beyond their immediate context windows.

To overcome this, Retrieval-Augmented Generation (RAG)—originally popularized in natural language processing [9]—has been adapted for time series. Current state-of-the-art approaches, such as TS-RAG [24] and TimeRAF [19], utilize deep neural encoders to project historical sequences into dense latent spaces, fusing retrieved context via learnable Adaptive Retrieval Mixers (ARM) or cross-attention mechanisms. While highly effective on dense, continuous datasets, these neural adapters break the fundamental promise of TSFMs: they require expensive pretraining on massive corpora and introduce millions of trainable parameters, rendering them computationally prohibitive for lightweight deployments.

Alternatively, parameter-free statistical retrieval (such as Euclidean k-Nearest Neighbors) offers zero-training overhead but suffers from a well-documented vulnerability: absolute scale mismatch. Because real-world time series exhibit non-stationarity and extreme magnitude variations, raw distance metrics frequently retrieve structurally dissimilar candidates, leading to catastrophic forecasting degradation.

In this work, we propose *ScaleRAG*, a universal mathematical framework that enables robust retrieval augmentation without a learned projector. By explicitly normalizing historical sequences prior to indexing and mathematically restoring the retrieved continuation to the query's magnitude prior to fusion, we eliminate the need for neural encoders and adaptive mixers in the retrieval path.

We are precise about where the framework is parameter-free. Indexing, nearest-neighbour search, and scale restoration introduce *zero* trainable parameters and require no pretraining: they are closed-form arithmetic over the query's own context statistics. Fusion of the restored continuation with the frozen backbone admits two variants. On dense continuous data (ETTm2) we use a fixed scalar weight, keeping the entire pipeline parameter-free. On the sparse M5 panel, where the value of retrieval varies sharply across series, we additionally employ a lightweight learned LightGBM gate over six diagnostic features (nearest-neighbour distance, retrieval disagreement, intermittency, log-volume, backbone uncertainty, and scale spread), fitted exclusively on training origins. This gate is the sole trainable component in the system and is three orders of magnitude smaller than the neural mixers it replaces; we report it explicitly rather than describing the M5 configuration as parameter-free.

Our contributions are threefold:

1. We empirically isolate the *Catastrophic Scale Mismatch* failure mode in non-neural retrieval, demonstrating that raw Euclidean k-NN degrades foundation model performance by a factor of $3.8\times$ on sparse retail data.
2. We introduce a deterministic, parameter-free scale restoration mechanism that recovers retrieval quality by $+85.4\%$ across modalities without requiring neural retraining.
3. We define a *Taxonomy of Retrieval Regimes*, showing that neural RAG models dominate dense, continuous datasets, whereas explicit statistical restoration is the viable augmentation strategy for sparse, zero-inflated domains and resource-constrained deployments where neural adapters risk scale hallucination or cannot be trained at all. We delimit this claim with a pre-registered held-out evaluation in which our method improves squared-error accuracy over its frozen backbone yet fails to clear the pre-registered margin over strong classical baselines, and we analyse why.

2 Related Work

2.1 Time Series Foundation Models

Prior to foundation models, deep learning for time series relied heavily on dataset-specific architectures, evolving from RNNs [16] and MLPs [14] to sophisticated Transformers [11, 13, 21, 25] and multi-scale mixers [18, 22]. Recently, the paradigm shifted toward zero-shot generalization. Large language models have been repurposed for forecasting [4, 6], while native time series models like Chronos [1] tokenize time series into discrete bins and train a language model to predict subsequent tokens, while models like Moirai [20] and TimesFM [3] employ patch-based attention mechanisms. Despite their zero-shot capabilities, these models are bounded by strict context window limits, restricting their ability to leverage long-term historical structural patterns.

2.2 Retrieval-Augmented Time Series

To ground foundation models in broader historical context, retrieval mechanisms have been proposed. TS-RAG [24] employs a neural encoder to fetch relevant historical sub-series, projecting them via a pre-trained Adaptive Retrieval Mixer (ARM). Similarly, TimeRAF [19] uses embedding similarity and neural fusion. While these methods achieve state-of-the-art accuracy on continuous benchmarks like ETT and Weather, they depend entirely on deep latent projections to circumvent numerical scale mismatch, necessitating computationally expensive retraining phases.

2.3 Non-Stationarity and Statistical Retrieval

The fragility of statistical similarity in non-stationary time series is widely recognized. Normalization techniques like Reversible Instance Normalization (RevIN) [8] attempt to mitigate distribution shifts internally within neural networks. On the retrieval front, SARAF [10] addresses this by modulating retrieval diversity based on regime shifts, while TRACE [2] abandons raw numerical similarity in favor of text-based multimodal alignment.

Closest to the mechanism we study is RAID [17], which likewise decomposes each retrieved trajectory into shape and scale, normalizing a neighbor by its own mean and standard deviation before mapping the aggregate back onto a target scale. We therefore do not claim that operation as novel, and we note that RAID also reports results on M5 and Favorita. The distinction lies in where the restoration statistics originate. RAID addresses true cold start, in which the query has no numeric history at all; its target mean and standard deviation consequently cannot be measured and are instead regressed by a trained perceptron from textual metadata, alongside a diffusion transformer and a gating network learned from scratch, with retrieval itself performed over frozen multilingual text embeddings. Our setting is the complement. An observed query context is available, so the same statistics are computed from it directly, exactly and at no parameter cost. The two approaches thus occupy disjoint regimes: one in which scale must be predicted, and one in which it can be measured.

Two further methods bound the claim from the other side. RAFT [5] retrieves over raw numeric windows using Pearson correlation, and it already performs a partial form of the operation we study: it treats the final value of each patch as an offset, subtracts it from the query and from every retrieved patch, and restores the query's offset to the output. That is location removal and target-location restoration, though no magnitude statistic is ever divided out, and RAFT trains its projections from scratch rather than augmenting a frozen backbone. Separately, kNN-MTS [23] augments a frozen pretrained forecaster with a retrieval branch that introduces no trainable parameters and fuses through the same convex blend we adopt, but it retrieves over learned embeddings and uses the retrieved futures unmodified, without any rescaling.

Taken together, the individual ingredients are therefore established: scale-insensitive numeric retrieval in RAFT, location restoration in RAFT, full mean and variance restoration of a retrieved trajectory in RAID, and parameter-free convex fusion onto a frozen backbone in kNN-MTS. What we do not find in this literature is their combination, namely a frozen backbone augmented by raw numeric retrieval whose retrieved future is restored in closed form using both location and scale measured directly from the query context. Our contribution is accordingly the composition and its empirical characterization rather than the operation in isolation, and we state it that way deliberately.

In contrast to the neural and semantic lines above, our approach directly confronts the numerical brittleness of Euclidean k-NN. Rather than bypassing raw similarity via latent embeddings or semantic text, we make the scale correction explicit and closed form. This yields a deterministic, computationally lightweight treatment of scale hallucination whose retrieval and restoration stages introduce no trainable parameters.

3 Methodology

To resolve the catastrophic scale mismatch inherent in raw Euclidean retrieval without introducing trainable neural adapters, we propose an explicit mathematical normalization and restoration pipeline, shown end to end in Fig. 1. The methodology consists of three deterministic phases: pre-retrieval scale normalization, scale-invariant neighbor search, and post-retrieval continuation restoration.

3.1 Pre-Retrieval Scale Normalization

Let $x \in \mathbb{R}^L$ denote a univariate time series context window of length L. To ensure the retrieval mechanism matches temporal shape rather than absolute magnitude, we strip the absolute scale from all historical candidate sequences and the target query sequence prior to indexing.

For each sequence x, we compute the context mean μ and the Root Mean Square (RMS) deviation σ_{RMS}:

$$\mu = \frac{1}{L}\sum_{t=1}^{L} x_t \tag{1}$$

$$\sigma_{\text{RMS}} = \sqrt{\frac{1}{L}\sum_{t=1}^{L}(x_t - \mu)^2} \tag{2}$$

The sequence is then normalized into a scale-invariant representation x^*:

$$x^* = \frac{x - \mu}{\sigma_{\text{RMS}} + \varepsilon} \tag{3}$$

where ε is a small numerical constant to prevent division by zero. All historical candidates are transformed into this normalized space and indexed using FAISS (Facebook AI Similarity Search) with an exact L_2 distance metric.

3.2 Scale-Invariant Nearest Neighbor Search

Given a normalized target query context $q^* \in \mathbb{R}^L$, the retriever identifies the top-k nearest neighbors from the indexed historical corpus. Because both the query and the historical corpus exist in the normalized space, the FAISS engine reliably identifies structural shape similarities irrespective of differing absolute sales volumes or sensor magnitudes.

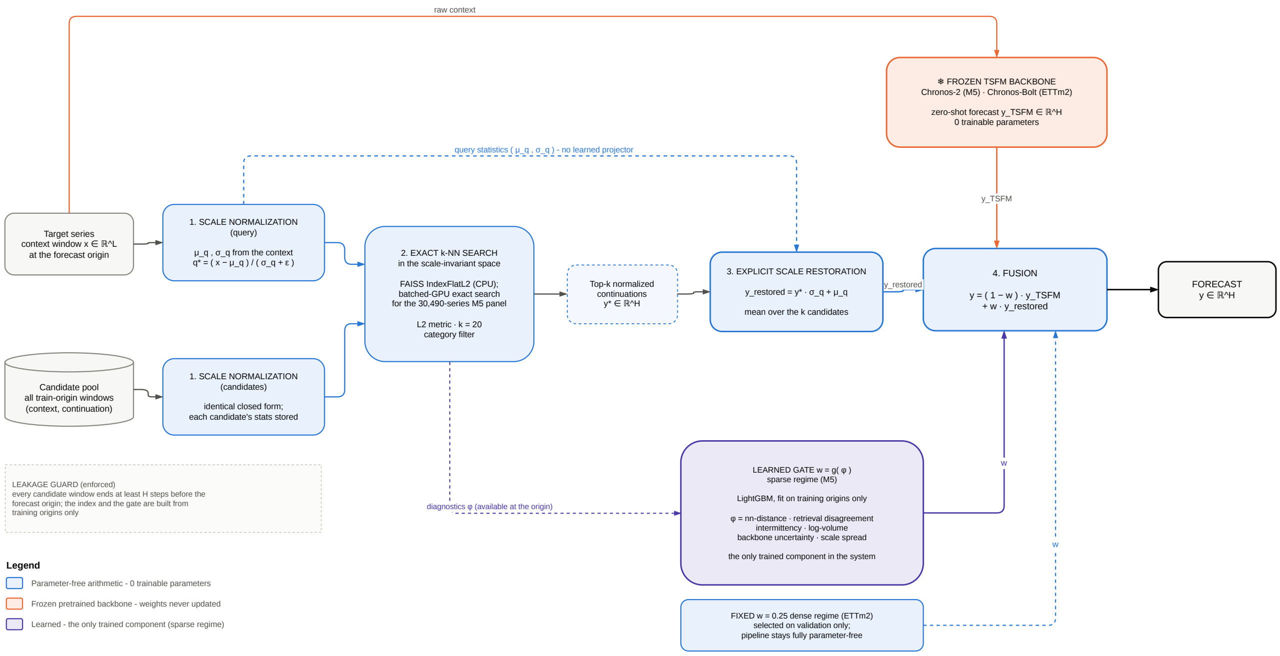

Fig. 1 The ScaleRAG pipeline. Normalization, search, and restoration (stages ①–③) are closed-form arithmetic and add no trainable parameters: the dashed path carries the query's own statistics (μ_q, σ_q) from normalization directly to restoration, which is what replaces the learned projector used by neural retrieval adapters. The backbone is frozen throughout, and the candidate pool is restricted to training origins with the horizon guard enforced, so no retrieved continuation can overlap the window it informs. Only the fusion weight w differs between regimes: a fixed constant on dense data, keeping the pipeline entirely parameter-free, or a lightweight LightGBM gate—the single trained component in the system—on the sparse M5 panel.

3.3 Explicit Scale Restoration of the Continuation

Each retrieved historical candidate consists of a normalized context history and a corresponding normalized future continuation, denoted as $y^*_{\text{retrieved}} \in \mathbb{R}^H$, where H is the forecast horizon.

Before fusing this historical future with the foundation model's zero-shot prediction, the continuation must be mathematically reconstructed to match the specific absolute magnitude of the current target query. Using the pre-calculated statistics of the query context (μ_q, σ_q), we perform explicit scale restoration:

$$y_{\text{restored}} = y^*_{\text{retrieved}} \cdot \sigma_q + \mu_q \tag{4}$$

This deterministic transformation guarantees that the retrieved continuation reflects the exact magnitude baseline of the current forecast origin, replacing the function of a learned neural projector with a zero-parameter mathematical formulation.

3.4 Fusion with the Frozen Backbone

The restored continuation y_{restored} is combined with the frozen backbone's zero-shot prediction $\hat{y}_{\text{TSFM}}$ by a convex weight $w \in [0, 1]$:

$$\hat{y} = (1 - w)\, \hat{y}_{\text{TSFM}} + w\, y_{\text{restored}} \tag{5}$$

We instantiate w in two ways, and the distinction is material to the parameter accounting reported in Sec. 4.

Fixed fusion (parameter-free). On dense, continuous data we fix w to a single constant selected on the validation split only ($w = 0.25$, $k = 20$, mean scaling). No component of the pipeline is trained, and the method inherits the backbone's zero-shot deployment profile exactly.

Gated fusion (one lightweight learned component). On sparse, intermittent panels the utility of retrieval varies strongly across series: for high-volume, regular items the frozen backbone is already close to optimal, whereas for zero-inflated items the restored continuation carries most of the signal. We therefore predict a per-series, per-origin weight $w = g(\phi)$ with a gradient-boosted regression tree (LightGBM [7]) over a six-dimensional diagnostic feature vector ϕ comprising the nearest-neighbour distance, the disagreement among the k restored continuations, an intermittency statistic, log-volume, the backbone's own predictive uncertainty, and the scale spread of the retrieved neighbours. Crucially, ϕ is computed entirely from information available at the forecast origin, and g is fitted on training origins only—never on validation or test origins—so no target information from the evaluation period enters the gate. This gate is the only trainable element of ScaleRAG.

4 Experiments and Results

To evaluate the efficacy of the proposed mathematical scale restoration, we conduct experiments across two distinct regimes: highly intermittent retail demand (M5) [12] and dense, continuous sensor data (ETTm2) [25]. We benchmark *ScaleRAG* against zero-shot target-only foundation models and state-of-the-art neural retrieval (TS-RAG).

All splits are strictly chronological. On M5, the held-out test window `d_1914--d_1941` was frozen before any test evaluation, opened exactly once, and never used to tune any component; the success criteria reported in Sec. 4.3 were pre-registered prior to that run. On ETTm2 the fused configuration (mean scaling, $k = 20$, $w = 0.25$) was selected on the validation split alone. Every retrieval candidate satisfies the horizon guard $t_r + H <$ origin, so no retrieved continuation can overlap the forecast window it informs.

4.1 The Scale Mismatch Ablation

Our primary hypothesis is that raw Euclidean retrieval on unscaled data suffers from a catastrophic magnitude mismatch, illustrated on a real query–candidate pair in Fig. 2. As shown in our evaluations, executing statistical k-NN on the sparse M5 dataset without scale restoration yields an unusable RMSSE of 2.7884. By explicitly reconstructing the retrieved continuation's magnitude, we prevent this catastrophic degradation, reducing RMSSE to 0.7425—a 3.8× recovery in predictive accuracy.

This mechanism transfers universally across modalities. On the dense ETTm2 benchmark, the restored retrieval candidate significantly outperformed the raw Euclidean candidate, yielding an +85.4% MSE improvement (95% CI: [+85.2, +85.6]) across all channels. Table 1 reports both regimes side by side, and Fig. 3 resolves the ETTm2 result per channel: all 7/7 channels improve, so the aggregate is not driven by a single outlier.

4.2 Regime Taxonomy and Boundary Conditions

While scale restoration definitively fixes statistical retrieval quality, its impact on the final end-to-end forecast is heavily regime-dependent. Fig. 5 plots the realised utility of retrieval augmentation against the zero-fraction of each dataset and makes the trend explicit: the benefit grows with sparsity and scale heterogeneity, and vanishes on dense continuous channels.

On the sparse, scale-heterogeneous M5 dataset, ScaleRAG improved the frozen Chronos-2 backbone by +5.49% RMSSE (paired-bootstrap 95% CI [+5.40%, +5.59%], full 30,490-series panel). In this regime, target-only models struggle with zero-inflation and severe intermittency, making the restored historical continuations valuable. Sec. 4.3 qualifies this number against the pre-registered criteria.

Conversely, on the continuous ETTm2 benchmark, the neural target-only backbone (Chronos-Bolt) is remarkably strong. While our restored candidates were structurally accurate (Fig. 4), fusing them with the backbone resulted in a marginal but statistically significant degradation (−0.85% MSE, 95% CI [−1.39%, −0.28%]) compared to the target-only baseline. Furthermore, the official TS-RAG neural adapter outperformed our statistical fusion by 2.30% MSE. Table 2 summarises the ETTm2 comparison.

4.3 Held-Out M5 Evaluation: Pre-Registered Criteria and Negative Findings

We report the held-out M5 outcome in full, including the results that do not favour our method. The +5.49% RMSSE gain over the frozen Chronos-2 backbone is real and its confidence interval excludes zero, but it does *not* generalise into a claim of state-of-the-art forecasting accuracy, for four reasons.

First, the margin over the strongest simple baseline is small. Against LightGBM [7], ScaleRAG improves RMSSE by only +0.69% (95% CI [+0.57%, +0.82%]). Our pre-registered success criterion required at least a 3% relative improvement over the strongest matched baseline, so this criterion **fails** despite the interval excluding zero. Statistical significance here is not practical significance: with 30,490 series, a sub-1% effect is comfortably detectable while remaining operationally negligible.

Second, ScaleRAG *loses the official M5 metric.* On WRMSSE—the dollar-weighted, hierarchical metric that defines the M5 competition—LightGBM (0.8663) and even Seasonal-Naive (0.8697) beat ScaleRAG (1.2231). Our method is mid-pack on the benchmark's own headline metric.

Third, the RMSSE gain does not carry over to absolute-error or probabilistic accuracy. The frozen target-only Chronos-2 backbone is the best model on MASE (0.893 vs. our 1.020), WAPE (0.665 vs. 0.698), MAE (0.960 vs. 1.008), pinball loss (0.289 vs. 0.298),

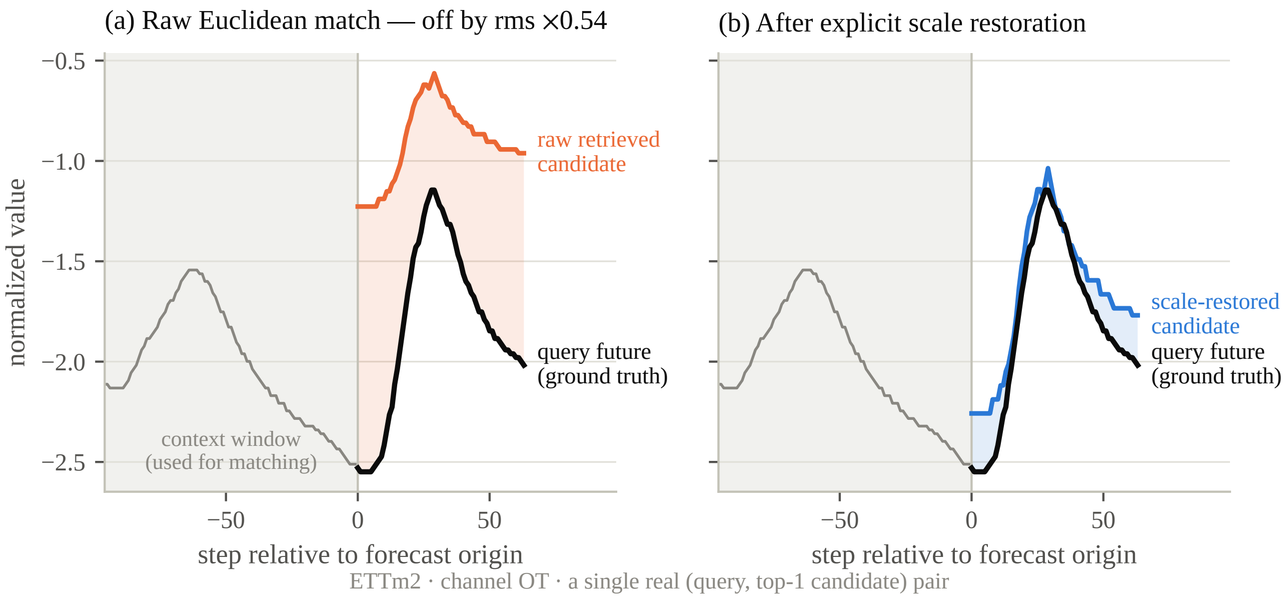

Fig. 2 The catastrophic scale-mismatch failure mode, on one real ETTm2 query and its actual top-1 Euclidean neighbour. The shaded band is the context window used for matching. (a) The retrieved candidate tracks the query's future *shape* closely but sits at an entirely different *magnitude*, so it is unusable as a forecast; the tint marks the residual gap. (b) Explicit restoration maps that same continuation onto the query's own scale using only the query context statistics—no learned projector, no trainable parameter.

Table 1 Scale restoration is decisive on both datasets. Raw retrieval normalises context windows for matching but does not rescale the retrieved continuation; restored retrieval additionally rescales candidate→query before use (Sec. 3). M5 reports RMSSE (lower is better, no CI available for this legacy ablation); ETTm2 reports MSE with a paired-bootstrap 95% CI (2,000 resamples, $n = 80{,}199$ windows, 7/7 channels improved). Relative improvement = (raw − restored)/raw; positive throughout.

Dataset	Metric	Raw retrieval	Restored retrieval	Relative improvement
M5 (1k, val)	RMSSE	2.7884	0.7425	+73.4% (3.8× lower)
ETTm2 (test)	MSE	2.9989	0.4376	+85.4% [+85.2%, +85.6%]

Table 2 Main results on ETTm2 (test, n = 80,199 windows). ΔMSE is the relative change vs. target-only Chronos-Bolt (positive = improvement). ScaleRAG's Δ values carry a paired-bootstrap 95% CI (window-level, $n = 80{,}199$); both CIs exclude zero. The TS-RAG Δ is its own reproduced point estimate (no CI computed here).

Method	MSE ↓	MAE ↓	ΔMSE vs. target-only
Chronos-Bolt (target-only, frozen)	0.1486	0.2235	—
TS-RAG (official ARM)	0.1465	0.2230	+1.41% (point estimate)
ScaleRAG (restored fixed fusion)	0.1498	0.2407	−0.85% [−1.39%, −0.28%]

and calibration: at a nominal 80% level, Chronos-2 attains 0.786 empirical coverage while our gated fusion under-covers at 0.698. Retrieval fusion therefore trades typical-case and distributional accuracy for reduced large squared errors on intermittent series.

Fourth, taken together, **0 of 3 pre-registered success criteria were met** on the held-out panel. We report this rather than re-tuning, because the test split was consumed exactly once by design and any post-hoc adjustment would invalidate it.

The honest reading is that no single method dominates: the winner reorders with the metric. This metric-dependent reordering, rather than a leaderboard position, is the empirical contribution—together with the mechanism-level result of Sec. 4.1, which is unambiguous and holds in both regimes.

4.4 Compute and Efficiency Trade-offs

TS-RAG achieves superior accuracy on continuous data but introduces 4.78M trainable parameters and significant VRAM overhead. ScaleRAG's retrieval and restoration stages introduce *zero* trainable parameters; on M5 the only trained component is the LightGBM gate

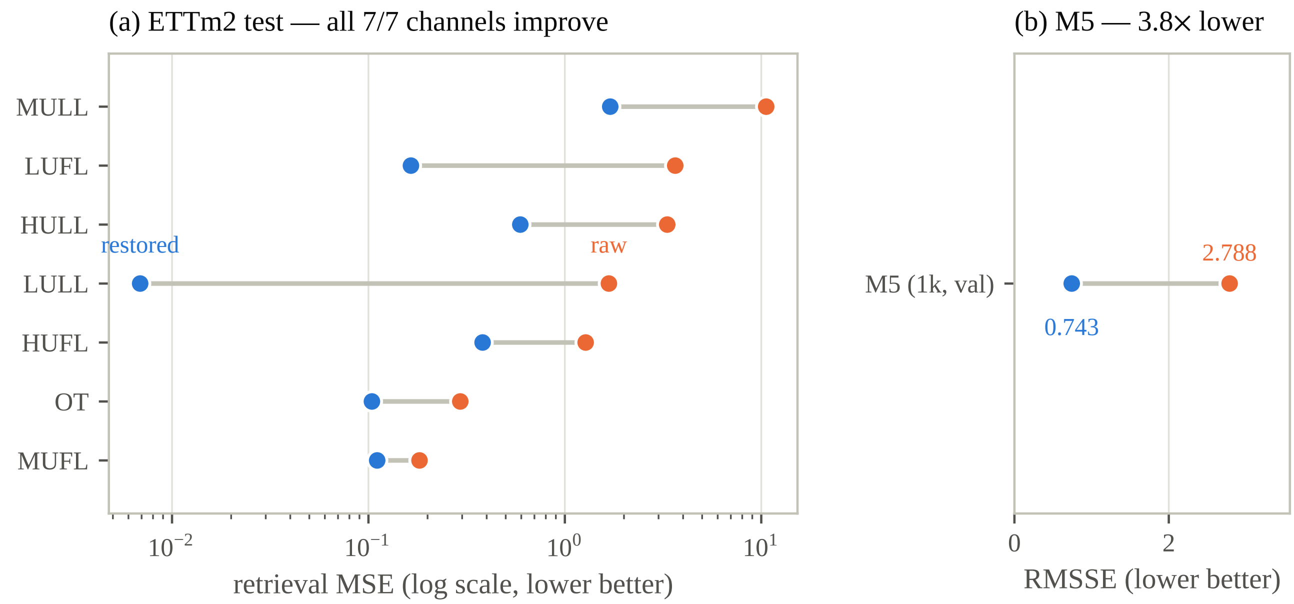

Fig. 3 Scale restoration moves retrieval error in the same direction in both regimes. Each row runs from raw retrieval (orange) to scale-restored retrieval (blue) for one population. (a) ETTm2 test split ($n = 80{,}199$ windows), per channel on a log scale: all 7/7 channels improve, so the aggregate +85.4% is not an artefact of averaging over a single dominant series. (b) M5 (1,000-series validation split), RMSSE: the same mechanism, a 3.8× reduction. The two panels use different metrics and are not comparable to each other—only within each row.

of Sec. 3.4, which is orders of magnitude smaller than a neural mixer and requires no pretraining corpus.

Search is *exact* in every configuration reported here—we never approximate the nearest-neighbour set, so accuracy differences cannot be attributed to index quantisation. Two exact implementations are used, matched to scale. The ETTm2 study uses FAISS with a flat CPU-resident `IndexFlatL2`, which is why Table 3 attributes 0 MB VRAM and 0.607 ms per window to the retrieval step. Scaling to the full 30,490-series M5 panel makes the CPU path intractable ($\mathcal{O}(n_{\text{query}} \times n_{\text{candidate}})$ at roughly five hours per origin), so for that panel we use a batched GPU implementation of the *same* exact search: the GPU performs a coarse float32 L_2 over-fetch, and the final top-k selection, tie-break, and float64 restoration are resolved exactly as in the CPU path. This GPU path was verified bit-for-bit identical to the frozen CPU retriever before any held-out run (maximum absolute difference 0.0); it is an arithmetic acceleration, not a method change.

Table 3 gives the full accounting and Fig. 6 plots the resulting accuracy–latency trade-off: on ETTm2, Scale-RAG is Pareto-dominated, being both slower end-to-end (1.038 ms vs. 0.450 ms per window) and less accurate than the official neural adapter. Fig. 7 shows that this conclusion is not an artefact of the neighbour budget: retrieval quality improves monotonically with k on both datasets (ETTm2 MSE 0.5362 → 0.4376; M5 RMSSE 0.7708 → 0.7425), and the fused forecast improves with it (0.1553 → 0.1498), yet remains above the target-only backbone (0.1486) at every $k \in \{5, 10, 20\}$.

Therefore, explicit statistical restoration is not proposed as a substitute for neural RAG on dense benchmarks, but rather as an operational option for sparse datasets and resource-constrained environments where neural adapters risk scale hallucination, cannot be pretrained, or are impossible to deploy.

5 Conclusion

In this paper, we identified and resolved the catastrophic scale mismatch inherent in non-neural retrieval for Time Series Foundation Models. By introducing a mathematical formulation that adds no trainable parameters to the retrieval path—normalizing context prior to indexing and explicitly restoring continuation magnitude post-retrieval using only the query's own statistics—we demonstrated an 85.4% recovery in retrieval MSE and prevented a 3.8× forecasting degradation on sparse data. Search remains exact throughout, executed by a flat CPU-resident FAISS index at benchmark scale and by a bit-identical batched GPU implementation of the same exact search on the full 30,490-series M5 panel.

We are deliberate about the parameter accounting. Indexing, search, and scale restoration are entirely parameter-free, and the dense-regime configuration

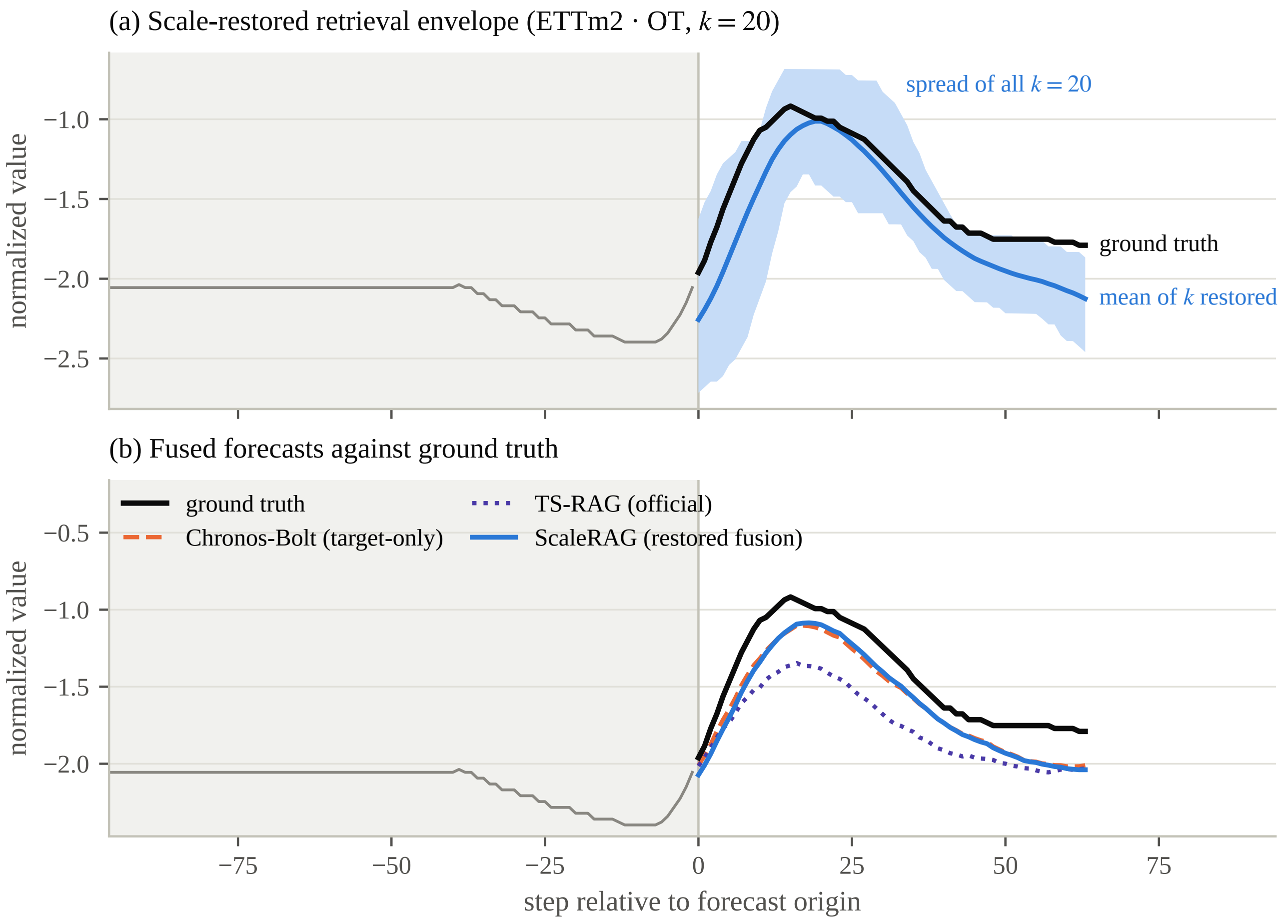

Fig. 4 A *representative* window on the ETTm2 OT channel (mean scaling, $k = 20$), chosen as the window whose fused error is closest to the median among windows with a non-trivial future—typical behaviour, not a best case. (a) The envelope spans all $k = 20$ scale-restored candidates, with their mean in blue: after restoration the retrieved set sits on the query's scale and reproduces the shape of the future, which is the mechanism the ablation measures. (b) The same window's forecasts. ScaleRAG's fusion tracks the frozen backbone closely and neither recovers the peak, illustrating how a genuine retrieval-quality gain can fail to convert into an end-to-end gain on a dense continuous channel.

Table 3 Measured compute, storage, and trainable-parameter accounting (ETTm2, one machine, batch 256, warm-up excluded). ScaleRAG's retrieval index is CPU-resident (0 VRAM); latency is reported separately for the frozen backbone forward pass and the retrieval step, since ScaleRAG's fused forecast requires both.

Component	Train. params	Latency / window	Peak VRAM	Peak RAM	Storage
Chronos-Bolt backbone (frozen)	0 (205.3M)	0.431 ms	1,562 MB	—	—
TS-RAG ARM (official)	4.78M	0.450 ms (+4.3%)	∼1.6 GB	—	1,528.6 MB*
ScaleRAG retrieval (non-neural)	0	0.607 ms (CPU)	0 (CPU)	1,245.0 MB	1,096.2 MB
ScaleRAG total (backbone + retrieval)	0	1.038 ms	1,562 MB	1,245.0 MB	—

*Full-series official knowledge base (not train-only); listed for storage reference, not a like-for-like index-size comparison.

keeps the full pipeline so. The sparse-regime configuration adds exactly one lightweight learned component—a LightGBM gate over six origin-available diagnostic features, fitted on training origins only—which remains three orders of magnitude smaller than the neural mixers it replaces, but which we decline to describe as parameter-free.

Crucially, our empirical analysis establishes a clear *Taxonomy of Retrieval Regimes*. State-of-the-art neural retrieval adapters excel on dense, continuous datasets, where their parameter overhead and retraining requirements buy real accuracy; on ETTm2 our statistical fusion is both slower and less accurate than the official neural adapter, and marginally worse than the frozen

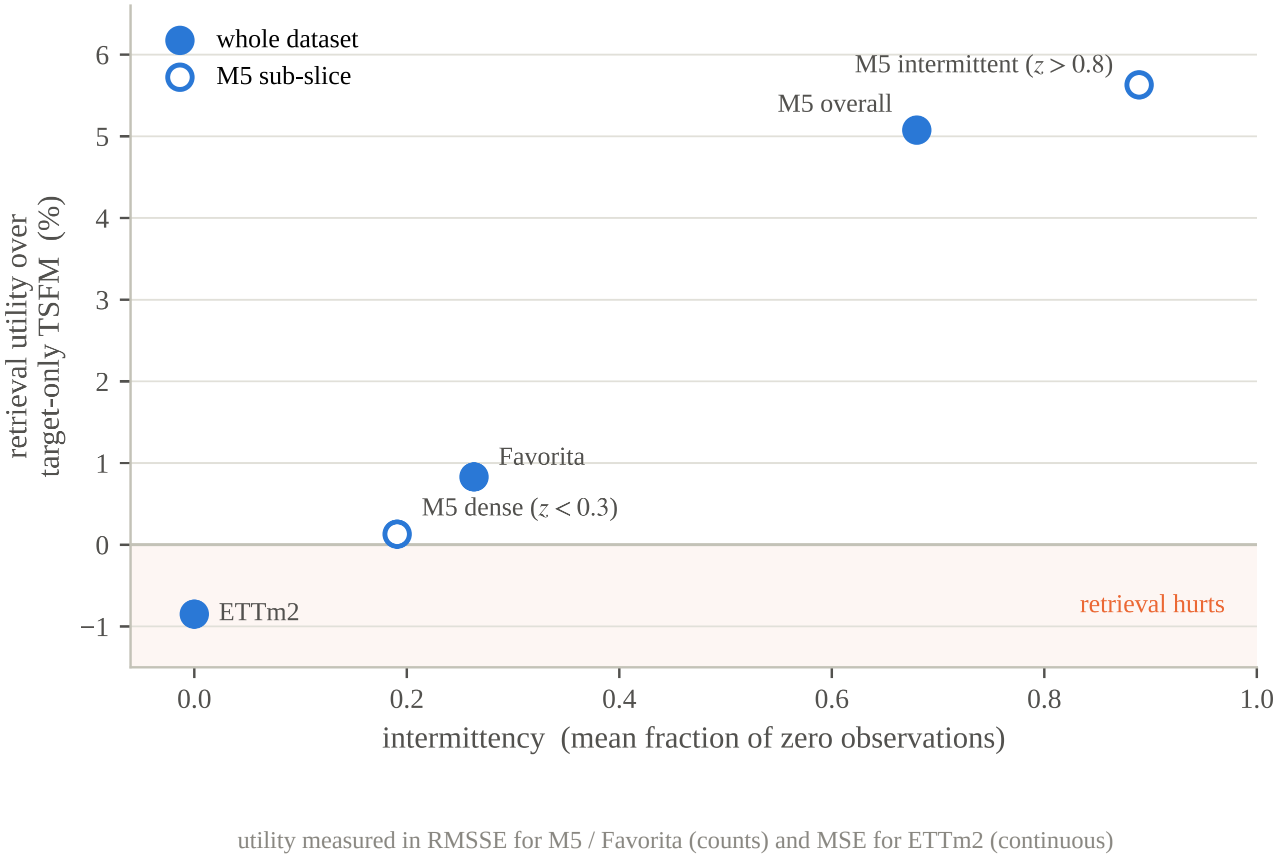

Fig. 5 Taxonomy of retrieval regimes. Realised utility of retrieval augmentation over the target-only foundation model, plotted against dataset zero-fraction (M5 and Favorita in RMSSE, ETTm2 in MSE). Retrieval augmentation pays off in sparse, scale-heterogeneous regimes and is neutral-to-harmful on dense continuous channels.

backbone it augments. For highly sparse, zero-inflated domains, or where resources preclude training a neural adapter at all, explicit mathematical scale restoration provides a robust, deployable augmentation strategy.

We also report the limits of that claim without qualification. On the held-out M5 panel, ScaleRAG improved squared-error accuracy over its frozen backbone by +5.49% RMSSE, yet beat the strongest classical baseline by only +0.69%—below our pre-registered 3% threshold—lost the official WRMSSE metric to both LightGBM and Seasonal-Naive, and was surpassed by the target-only backbone on MAE, WAPE, MASE, pinball loss, and interval coverage. All three pre-registered success criteria failed. The contribution of this work is therefore a validated mechanism and a regime taxonomy: scale restoration is demonstrably necessary for statistical retrieval to function at all, while the end-to-end value of that retrieval is bounded by regime and by metric. We regard the delineation of that boundary as more useful to practitioners than a leaderboard result, and we leave calibration-preserving fusion—the clearest deficiency our evaluation exposed—to future work.

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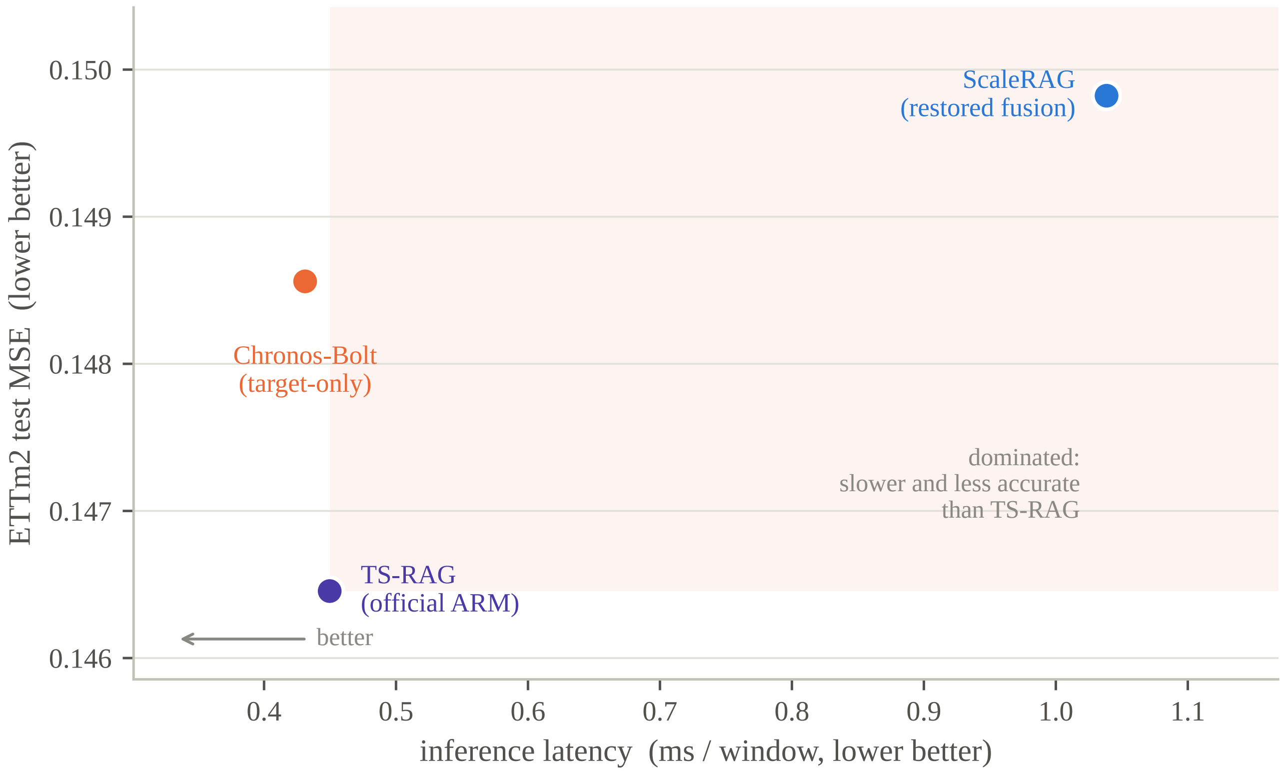

Fig. 6 Accuracy versus latency per window on ETTm2. ScaleRAG is Pareto-dominated in this regime: the official TS-RAG adapter is both faster end-to-end and more accurate. The efficiency argument for ScaleRAG rests on trainable-parameter count and deployability, not on wall-clock latency.

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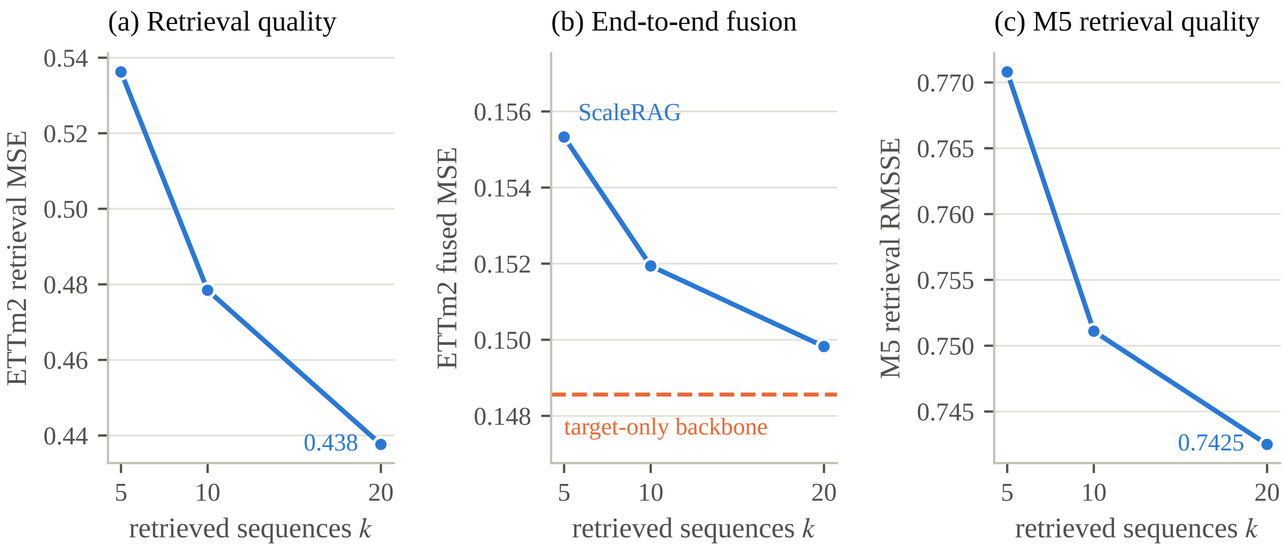

Fig. 7 Sensitivity to the neighbour budget k (mean scaling, fusion weight 0.25), shown as three small multiples because the three quantities differ in scale and two of them are different metrics—never as a twin axis. Retrieval quality improves monotonically with k on ETTm2 (a) and M5 (c), and the fused forecast improves too (b), but stays above the target-only backbone (dashed) at every k. The negative end-to-end result is therefore not an artefact of the frozen $k = 20$ choice.